

Choice D is the best answer because it describes data in the graph that support Meng and colleagues' conclusion that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T. According to the text, payload capacity is calculated by using a ratio of a robot's holding force to the robot's weight, and higher ratios indicate a greater payload capacity. The Ultra-Fast Robot Hand has a holding force of 56 newtons, four times greater than that of the Yale Model T. Additionally, the graph shows that the Ultra-Fast Robot Hand has a weight of approximately 500 grams, slightly more than the Yale Model T's weight of approximately 480 grams. Therefore, the Ultra-Fast Robot Hand has a higher ratio of holding force to weight than the Yale Model T. Since higher ratios correspond to greater payload capacity, the information from the graph indicating that the Ultra-Fast Robot Hand weighs only slightly more than the Yale Model T combined with the information in the text ultimately supports the conclusion that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T.

Choice A is incorrect. Although, according to the graph, it's true that both the Ultra-Fast Robot Hand and the Yale Model T weigh more than 450 grams, this statement doesn't support Meng and colleagues' conclusion that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T. This statement emphasizes a similarity, not a distinction, between the two robots. Choice B is incorrect. Although, according to the graph, it's true that the Ultra-Fast Robot Hand and the Yale Model T both weigh more than the Permanent Magnet Hand does, this statement doesn't support Meng and colleagues' conclusion that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T. This statement emphasizes a similarity, not a distinction, between the Ultra-Fast Robot Hand and the Yale Model T. Furthermore, the comparison to the Permanent Magnet Hand is irrelevant to the claim about the relative ratios and payload capacities of the Ultra-Fast Robot Hand and the Yale Model T. Choice C is incorrect. Although the text states that the Yale Model T has a lower holding force than the Permanent Magnet Hand, the graph provides no information about holding force. Moreover, information about the Permanent Magnet Hand is irrelevant to the conclusion by Meng and colleagues, which only concerns the Ultra-Fast Robot Hand and the Yale Model T.